CS535: Algorithm Design and Analysis

Red-Black Trees

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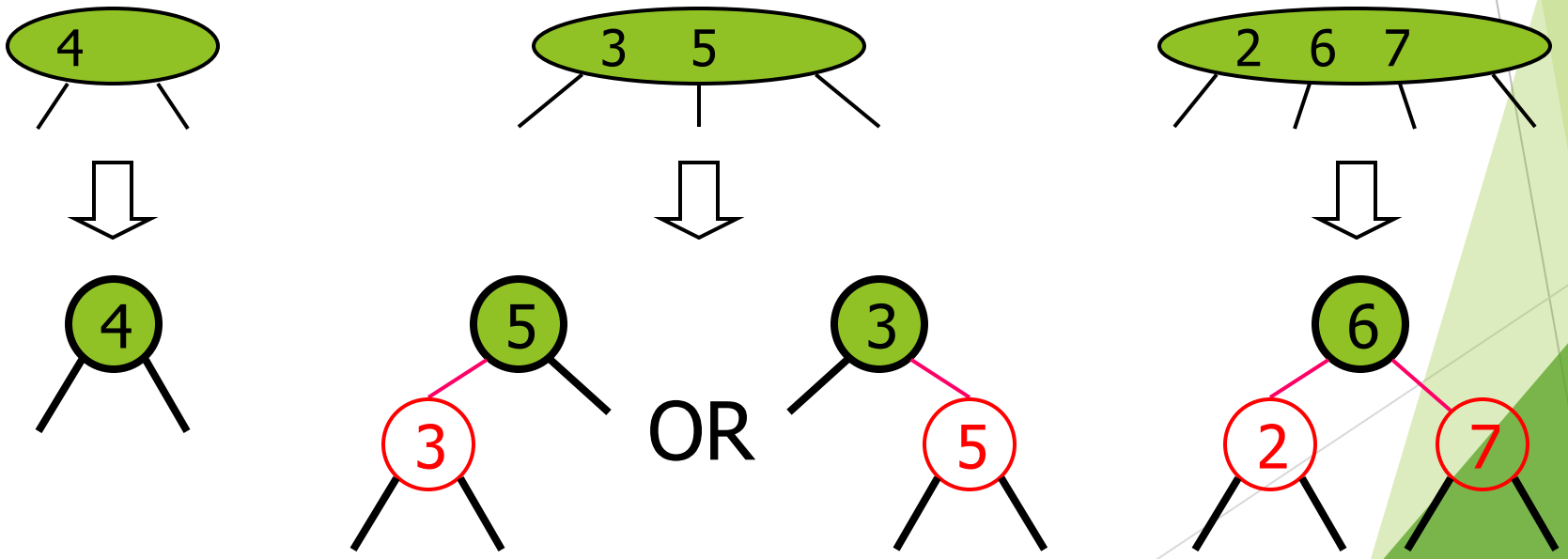
Readings: Chapter 3.3

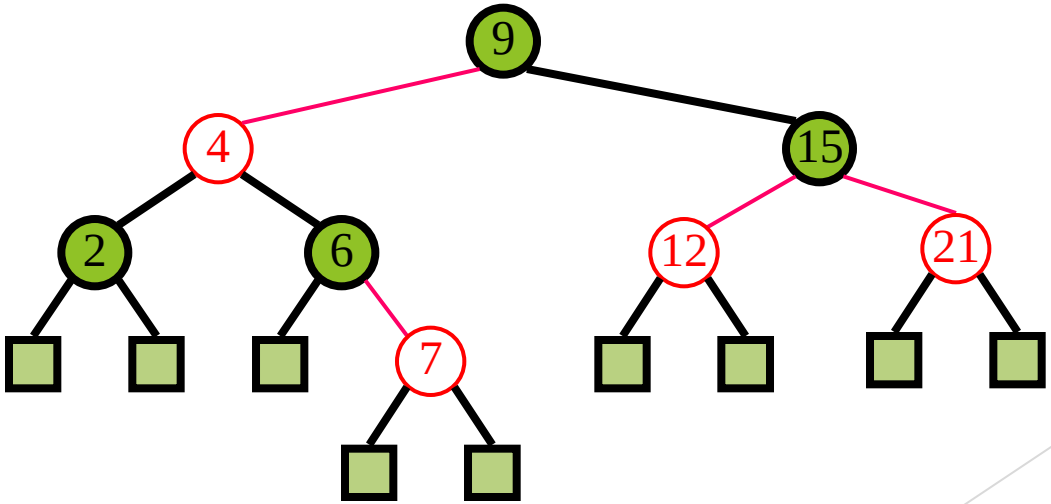
From (2,4) to Red-Black Trees

- A red-black tree is a representation of a (2,4) tree by means of a binary tree whose nodes are colored **red** or **black**
- In comparison with its associated (2,4) tree, a red-black tree has
  - same logarithmic time performance
  - simpler implementation with a single node type



Red-Black Tree

- A red-black tree can also be defined as a binary search tree that satisfies the following properties:
  - **Root Property**: the root is black
  - **External Property**: every leaf is black
  - **Internal Property**: the children of a red node are black
  - **Depth Property**: all the leaves have the same black depth



Height of a Red-Black Tree

- Theorem: A red-black tree storing n items has height $O(\log n)$

  Proof:

  - The height of a red-black tree is at most twice the height of its associated (2,4) tree, which is $O(\log n)$

- The search algorithm for a red-black tree is the same as that for a binary search tree

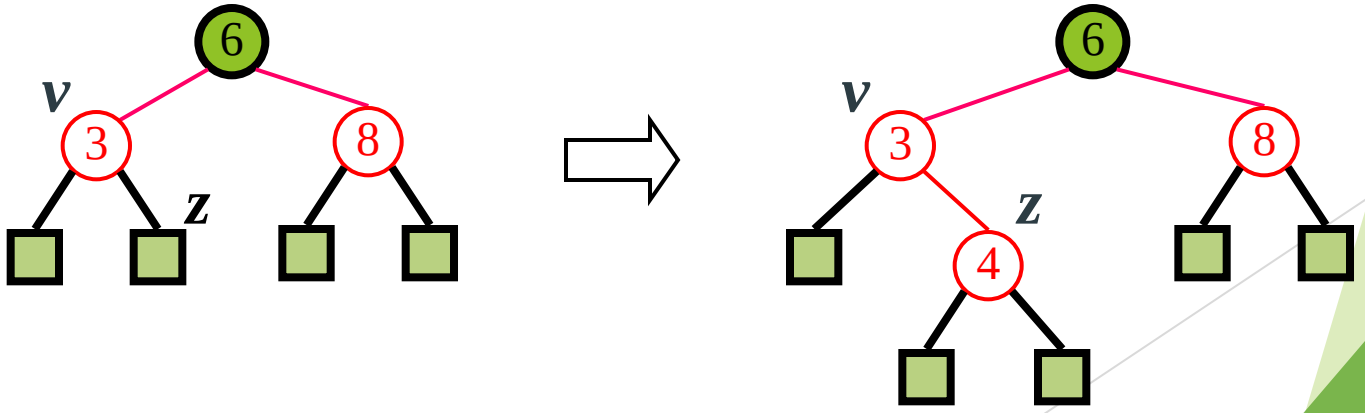
- By the above theorem, searching in a red-black tree takes

  $O(\log n)$ time

Insertion

- To perform **insertItem** (k, o), we execute the insertion algorithm for binary search trees and color red the newly inserted node z unless it is the root
  - We preserve the root, external, and depth properties
  - If the parent v of z is black, we also preserve the internal property and we are done
  - Else (v is red) we have a double red (i.e., a violation of the internal property), which requires a reorganization of the tree

- Example where the insertion of 4 causes a double red:



Remedying a Double Red

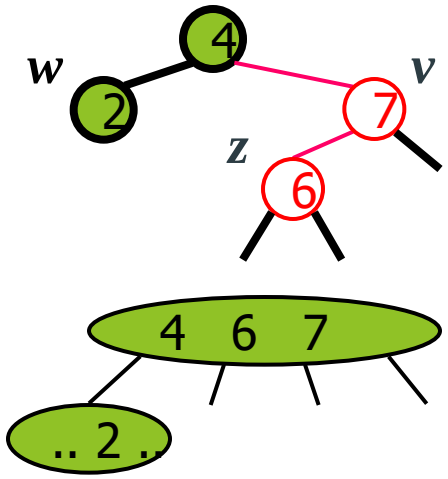
- Consider a double red with child z and parent v, and let w be the sibling of v

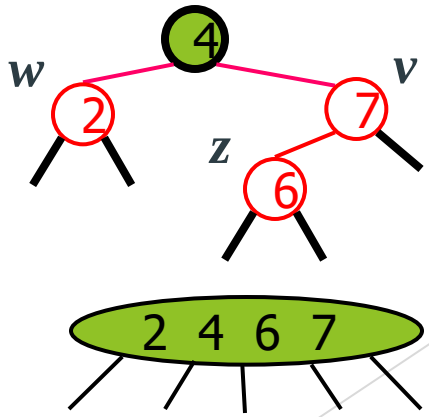
Case 1: w is black

- The double red is an incorrect replacement of a 4-node
- **Restructuring**: we change the 4-node replacement

Case 2: w is red

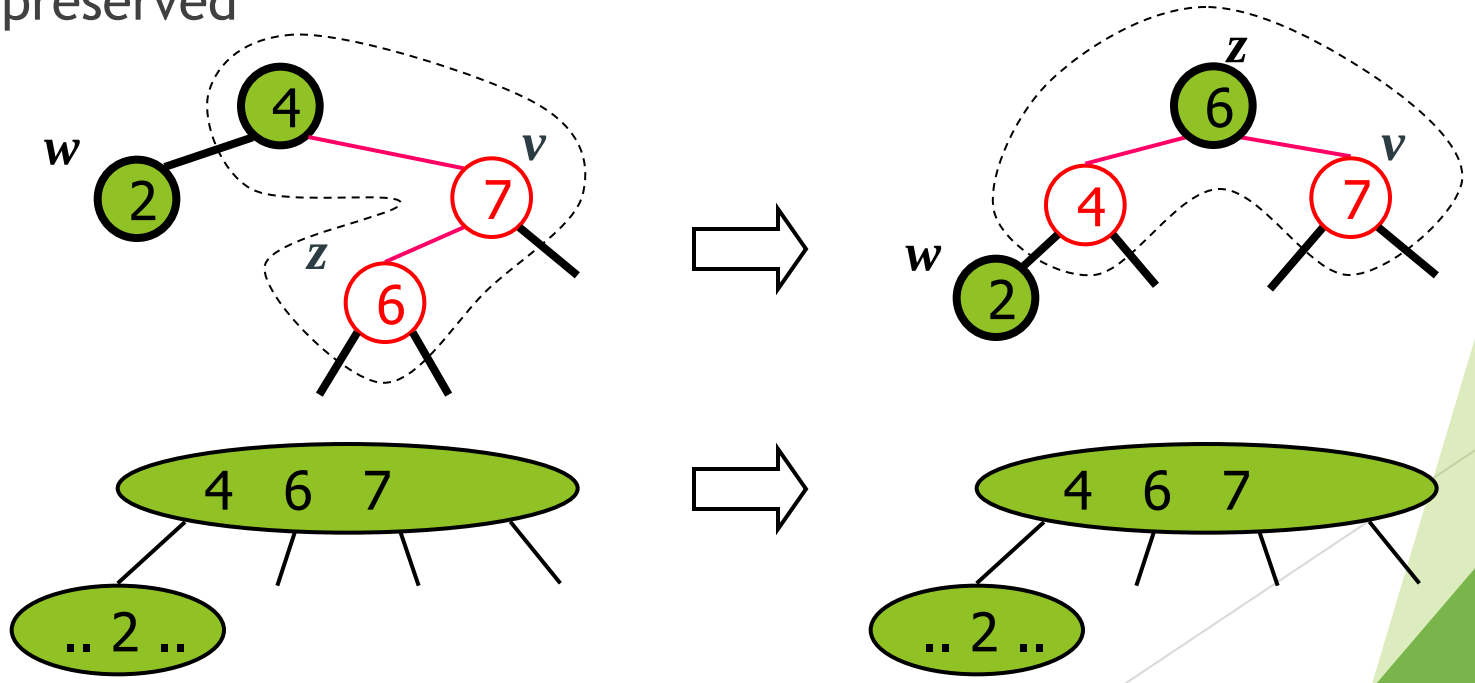
- The double red corresponds to an overflow
- **Recoloring**: we perform the equivalent of a split





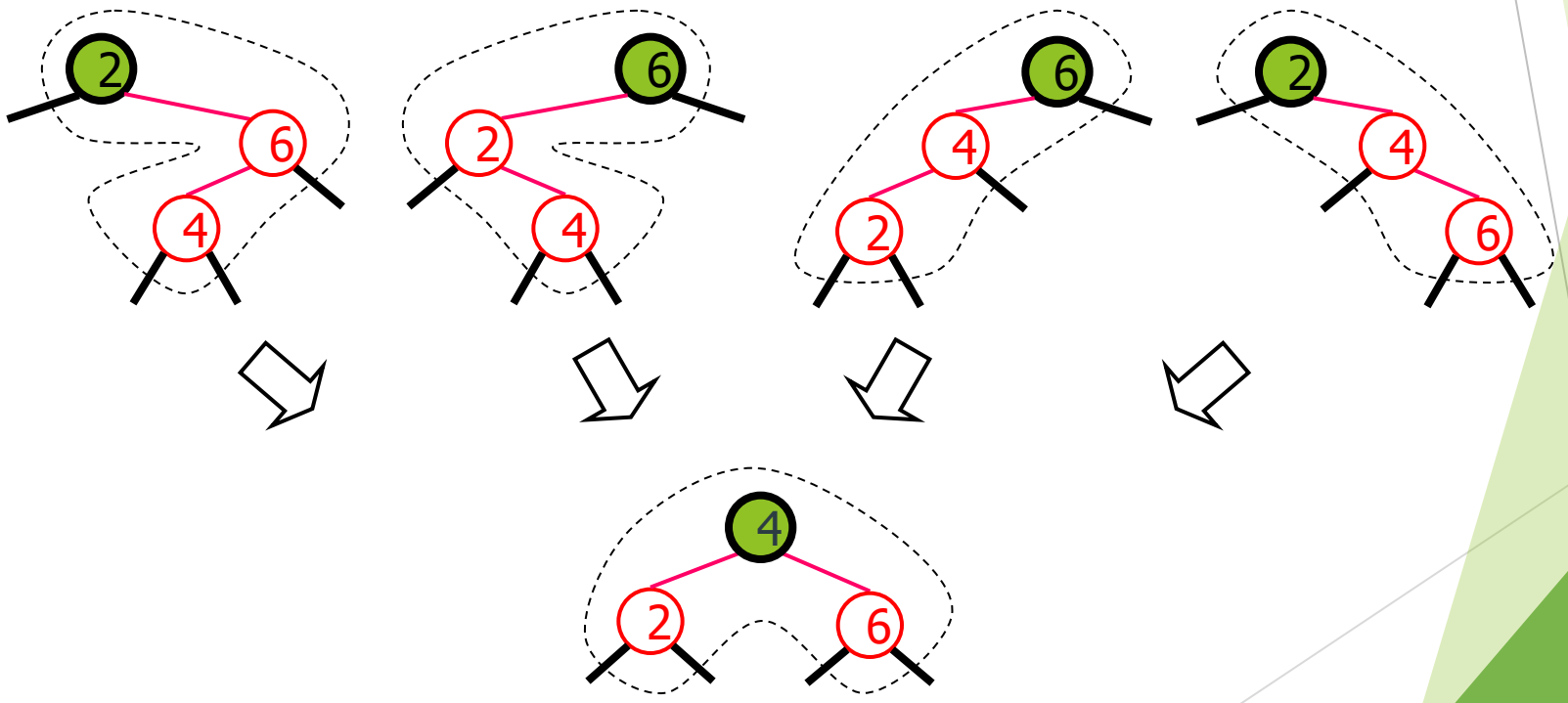
Case 1: Restructuring

- A restructuring remedies a child-parent double red when the parent red node has a black sibling
- It is equivalent to restoring the correct replacement of a 4-node
- The internal property is restored and the other properties are preserved



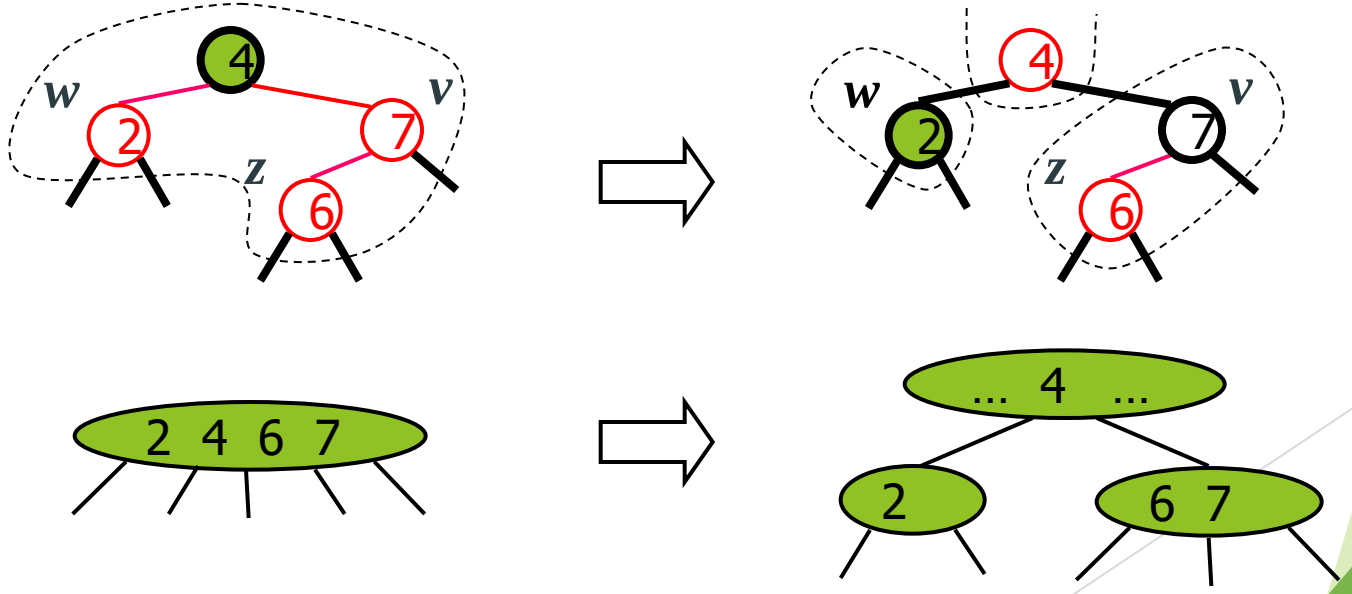
Case 1: Restructuring (cont.)

- There are four restructuring configurations depending on whether the double red nodes are left or right children



Case 2: Recoloring

- A recoloring remedies a child-parent double red when the parent red node has a red sibling
- The parent v and its sibling w become black and the grandparent u becomes red, unless it is the root
- It is equivalent to performing a split on a 5-node
- The double red violation may propagate to the grandparent u



Analysis of Insertion

Algorithm ***insertItem***(k, o)

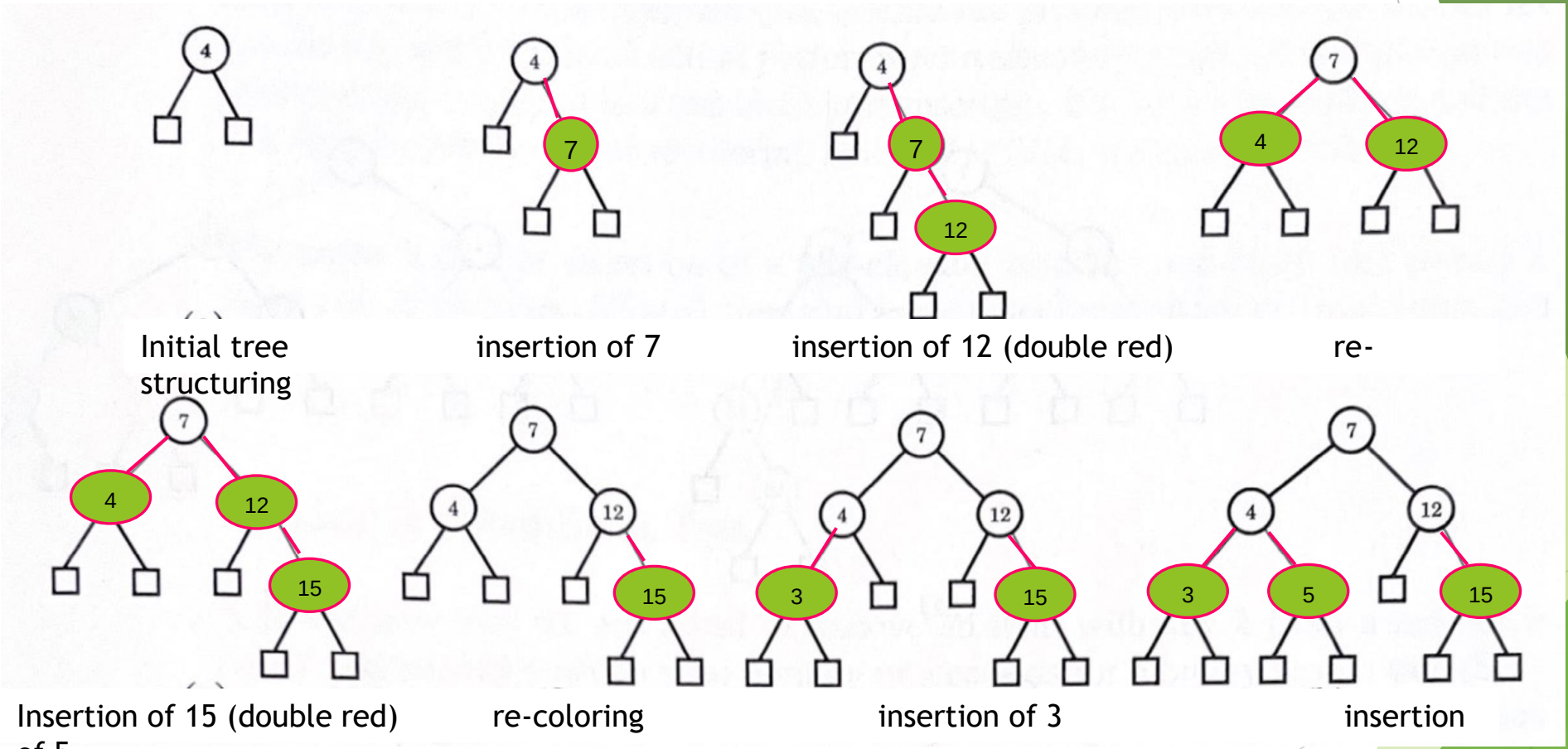
1. We search for key k to locate the insertion node z

2. We add the new item (k, o) at node z and color z red

3. **while** ***doubleRed***(z)
   if ***isBlack***(***sibling***(***parent***(z)))
   $z \leftarrow$ ***restructure***(z)
   return
   else { ***sibling***(***parent***(z) is red }
   $z \leftarrow$ ***recolor***(z)

- Recall that a red-black tree has $O(\log n)$ height
- Step 1 takes $O(\log n)$ time because we visit $O(\log n)$ nodes
- Step 2 takes $O(1)$ time
- Step 3 takes $O(\log n)$ time because we perform
  - $O(\log n)$ recolorings, each taking $O(1)$ time, and
  - at most one restructuring taking $O(1)$ time
- Thus, an insertion in a red-black tree takes $O(\log n)$ time

Example: A Sequence of Insertions



Initial tree

insertion of 7

insertion of 12 (double red)

re-structuring

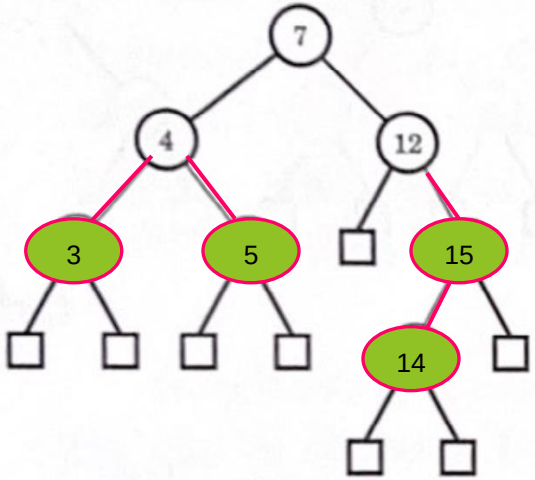
Insertion of 15 (double red)

re-coloring

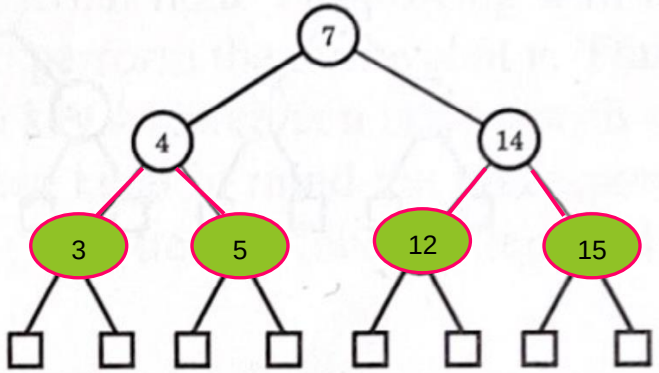
insertion of 3

insertion of 5

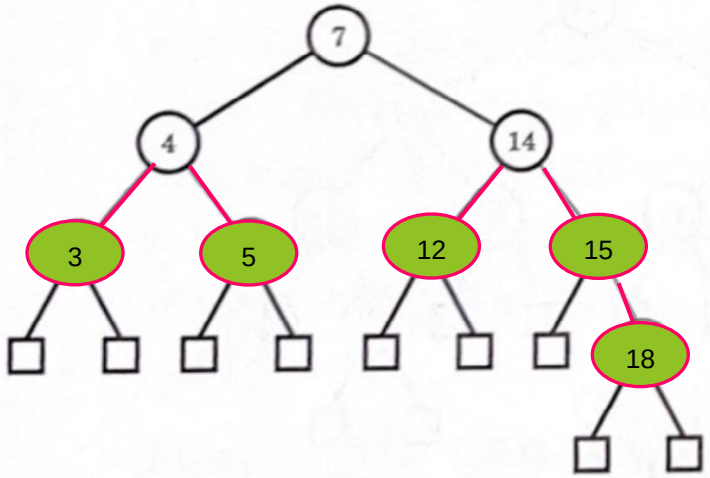
Example: A Sequence of Insertions



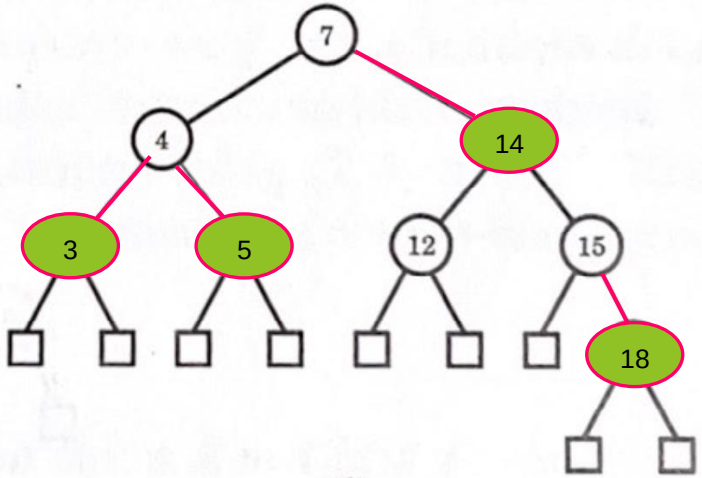
insertion of 14 (double red)



re-structuring

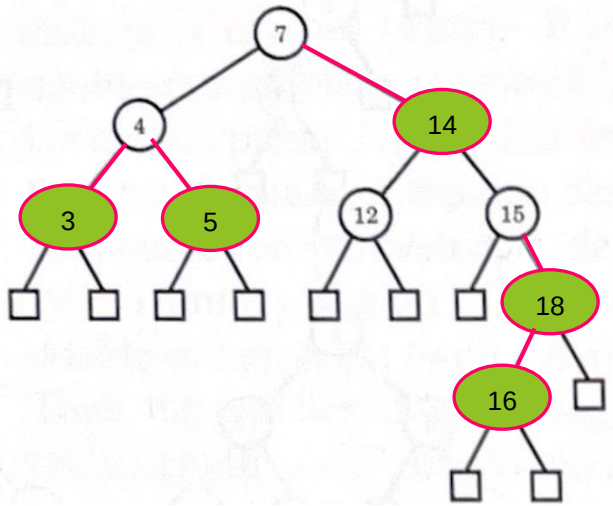


insertion of 18 (double red)

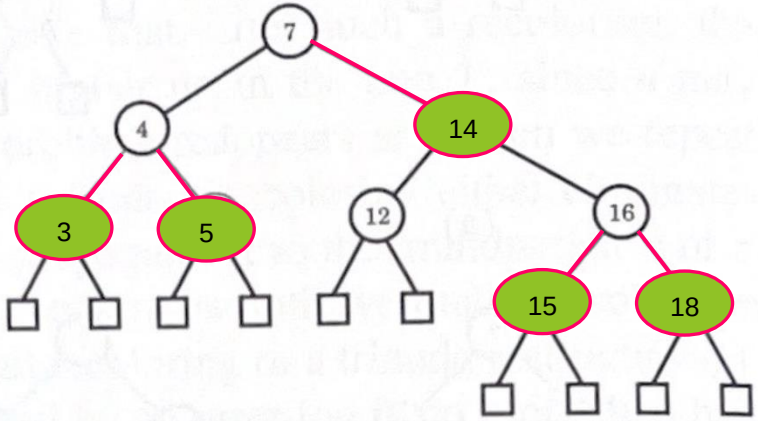


re-coloring

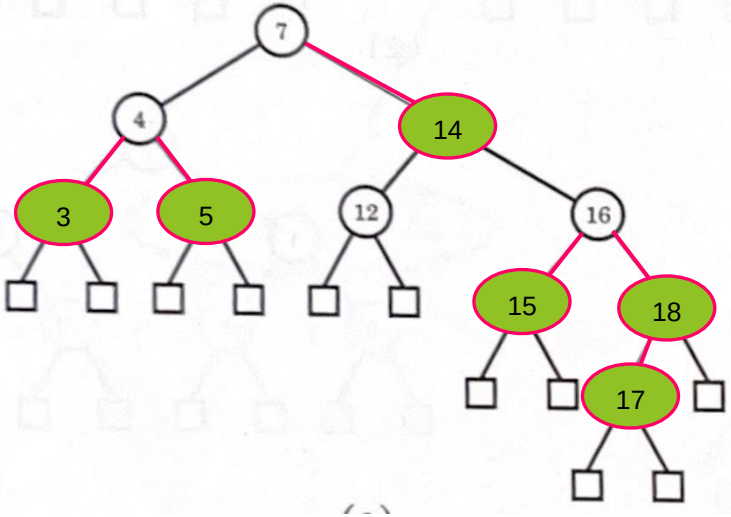
Example: A Sequence of Insertions



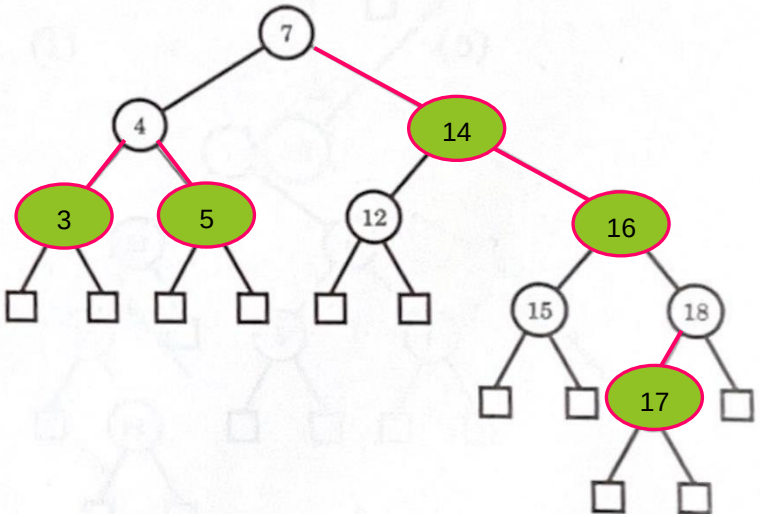
insertion of 16 (double red)



re-structuring

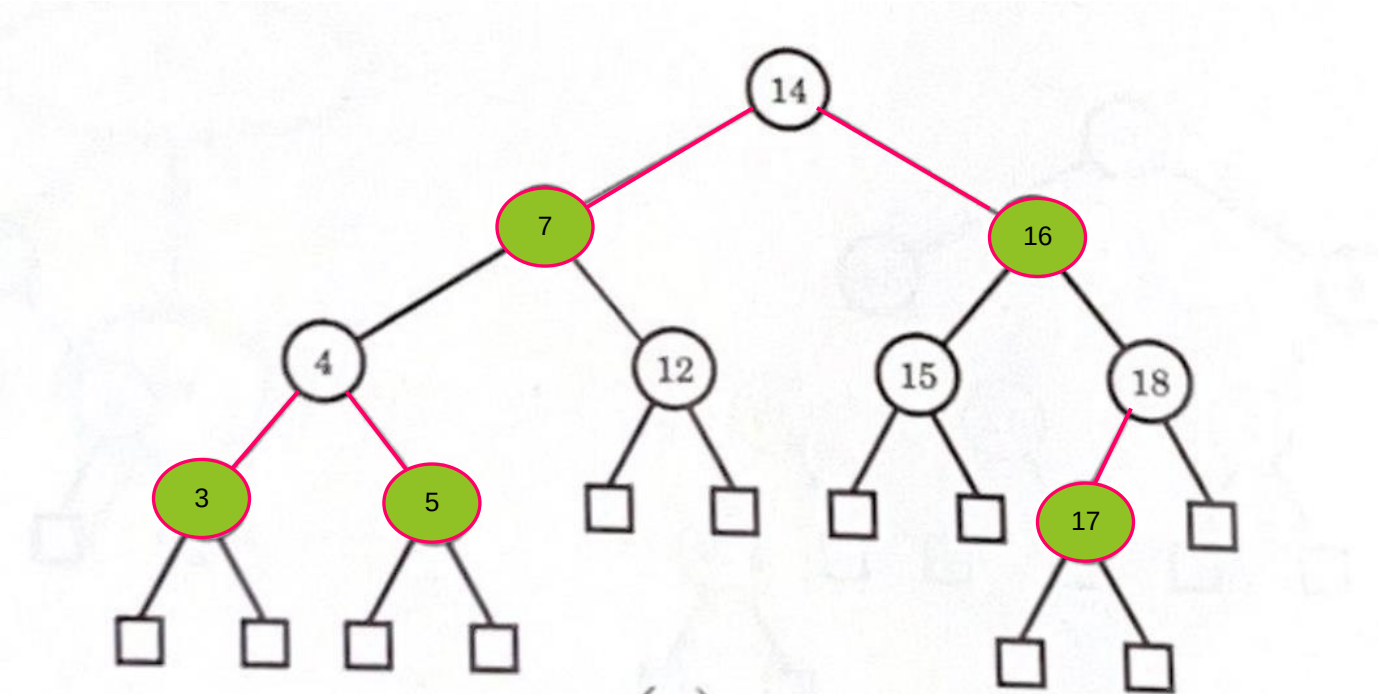


insertion of 17 (double red)



re-coloring

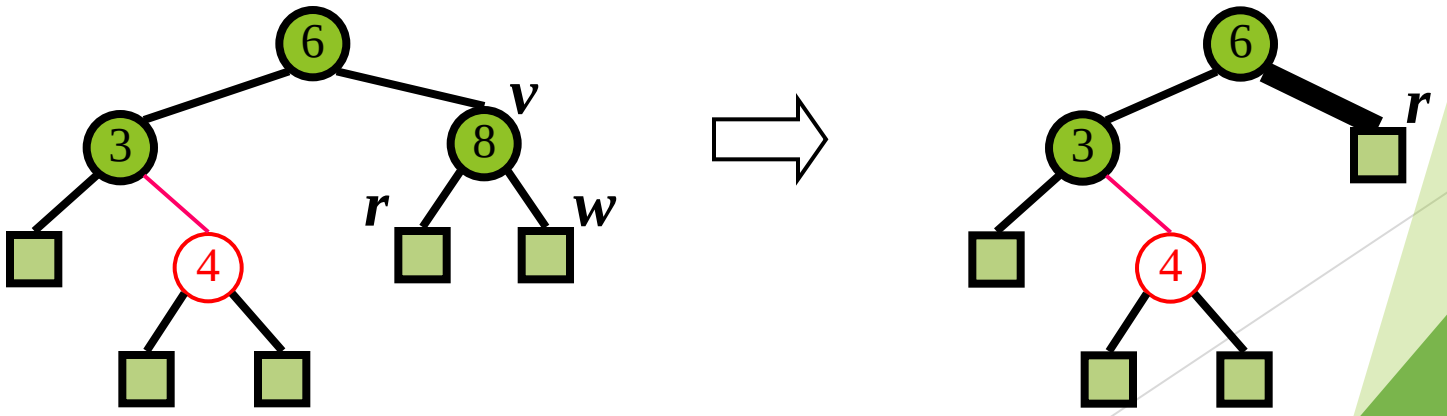
Example: A Sequence of Insertions



re-structuring

Removal

- To perform **remove** (k), we first execute the deletion algorithm for binary search trees
- Let v be the internal node removed, w the external node removed, and r the sibling of w
  - If either v or r was red, we color r black and we are done
  - Else (v and r were both black) we color r ***double black***, which is a violation of the depth property requiring a reorganization of the tree
- Example where the deletion of 8 causes a double black:



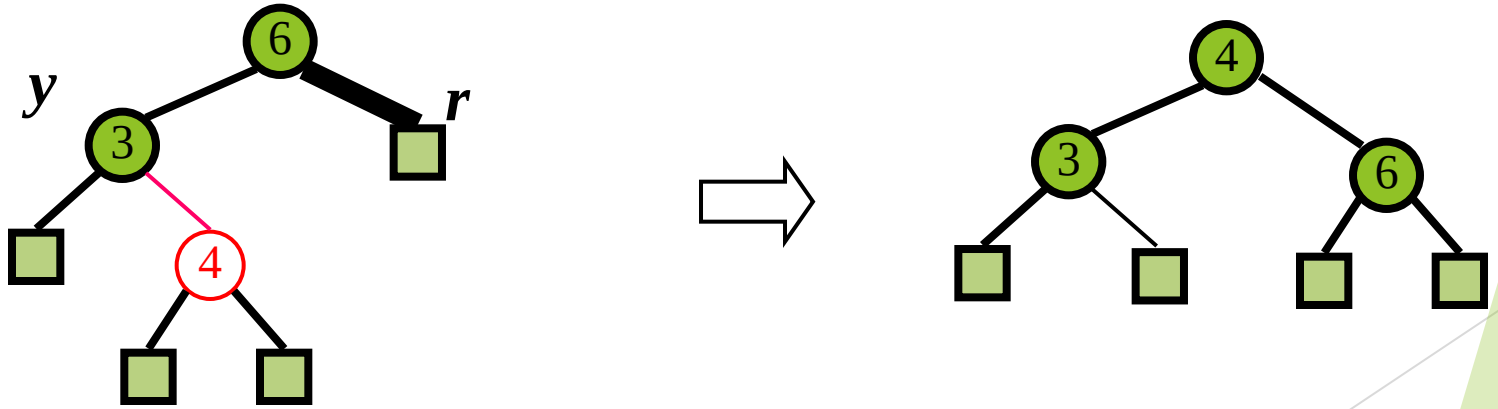
Remedying a Double Black

- The algorithm for remedying a *double black* node r with sibling y considers three cases

  Case 1: y is black and has a red child

  - We perform a **restructuring**, equivalent to a transfer , and we are done

    (in the example, move 4 up and 3 and 6 become the children. The red child disappears)



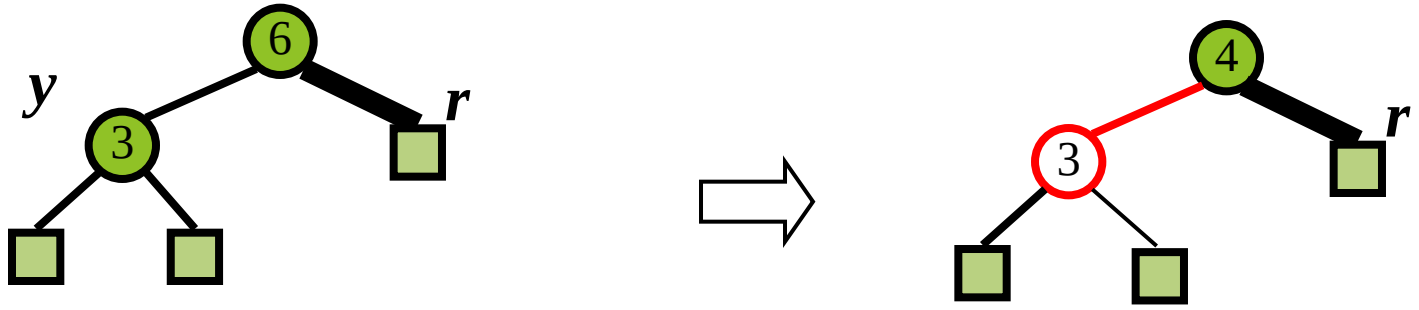
Remedying a Double Black

- The algorithm for remedying a *double black* node r with sibling y considers three cases

  Case 2: y is black and its children are both black

  - We perform a **recoloring**, equivalent to a fusion, which may propagate up the double black violation

    (in the example, color the node 3 and associated edge as red. The number of black nodes in this subtree may be smaller by one than its sibling)

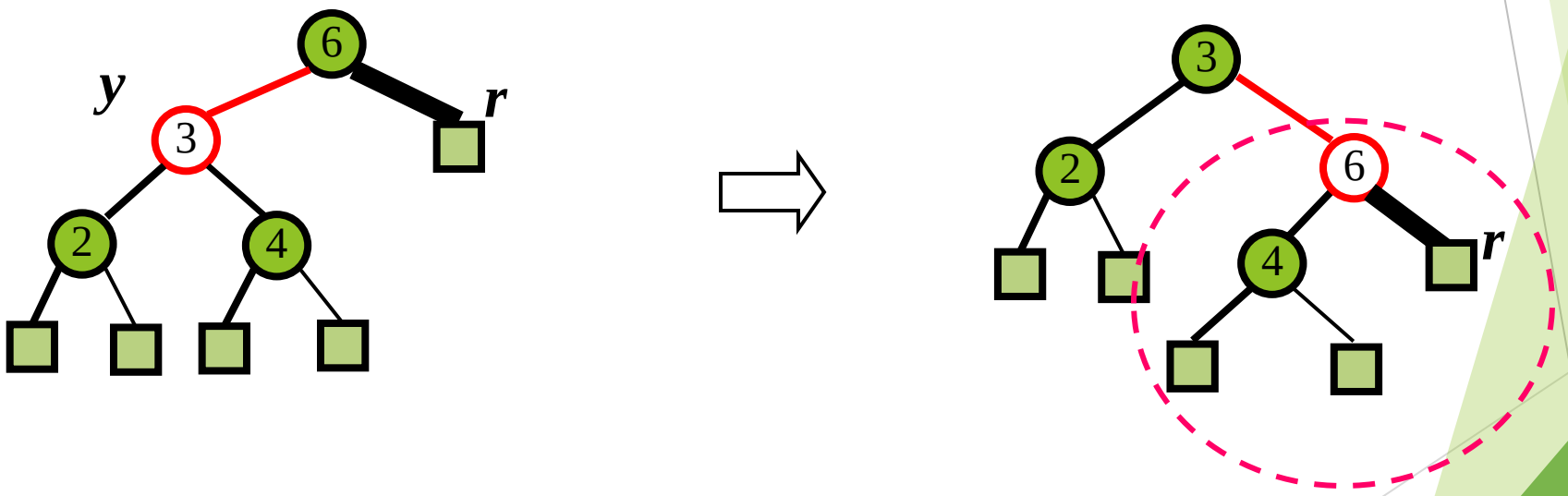


Remedying a Double Black

- The algorithm for remedying a *double black* node r with sibling y considers three cases

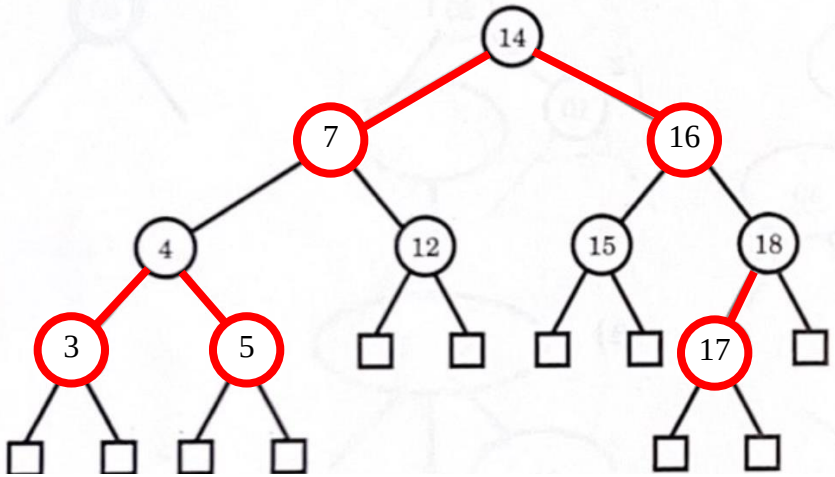
  Case 3: y is red

  - We perform an **adjustment**, equivalent to choosing a different representation of a 3-node, after which either Case 1 or Case 2 applies

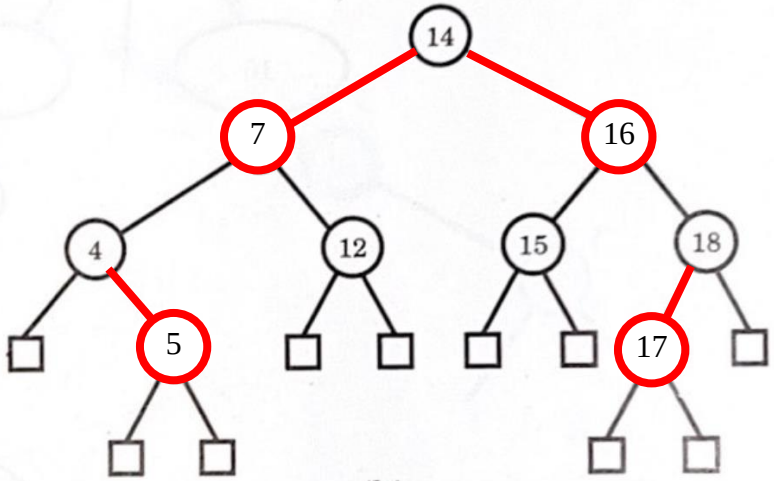


- Deletion in a red-black tree takes $O(\log n)$ time

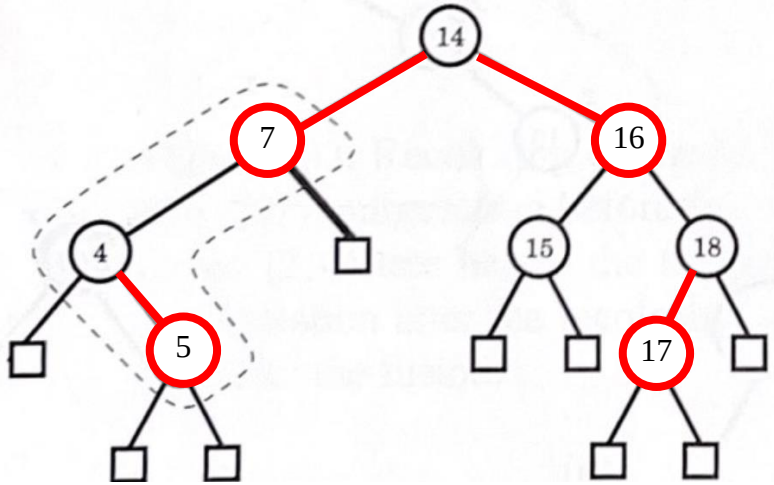
Example: A Sequence of Removals



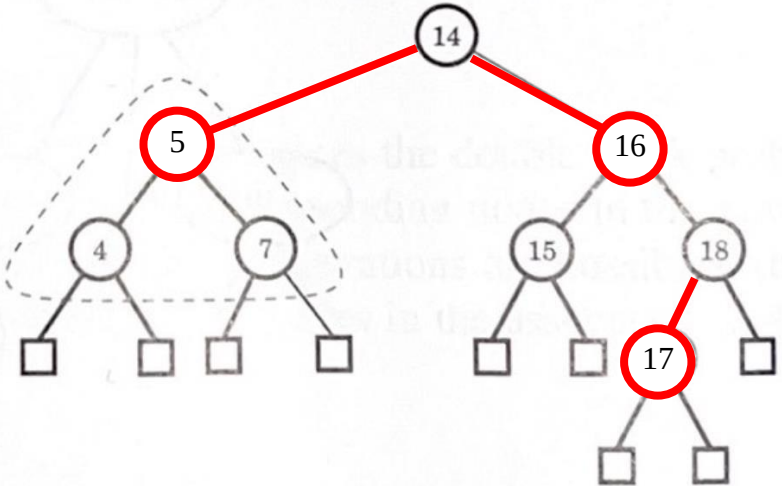
Initial tree



Removal of 3

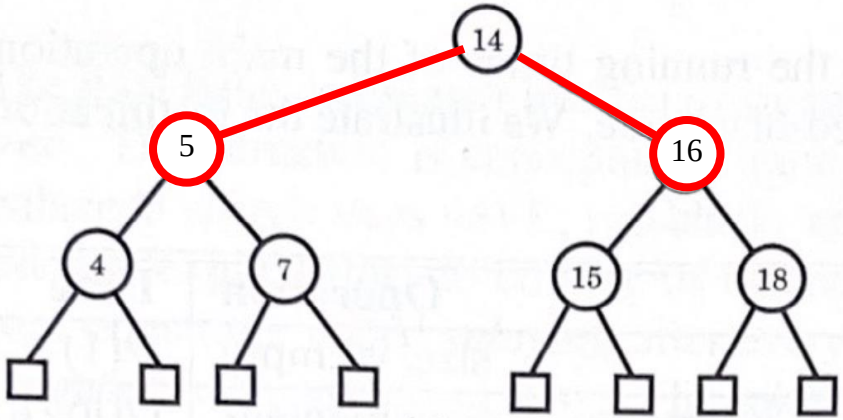


Removal of 12 (double black)

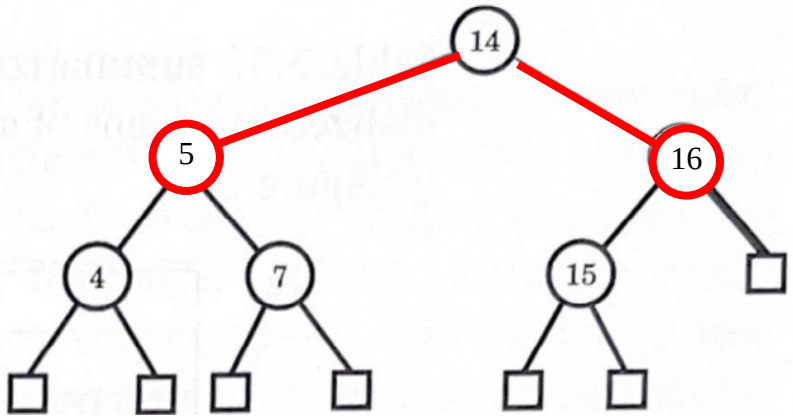


Re-structuring

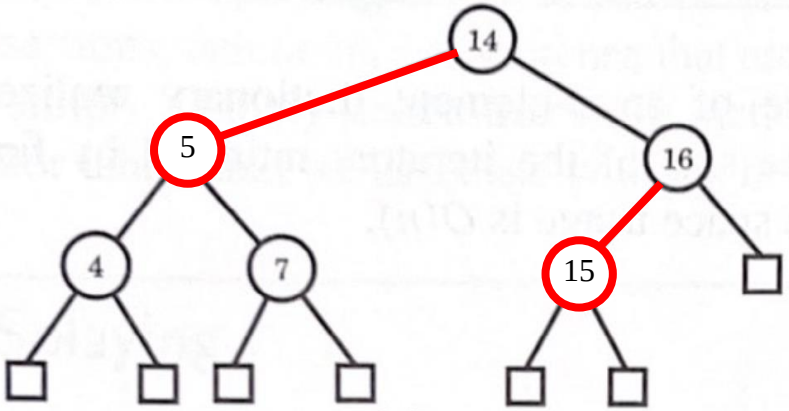
Red-Black Trees: Removal Example



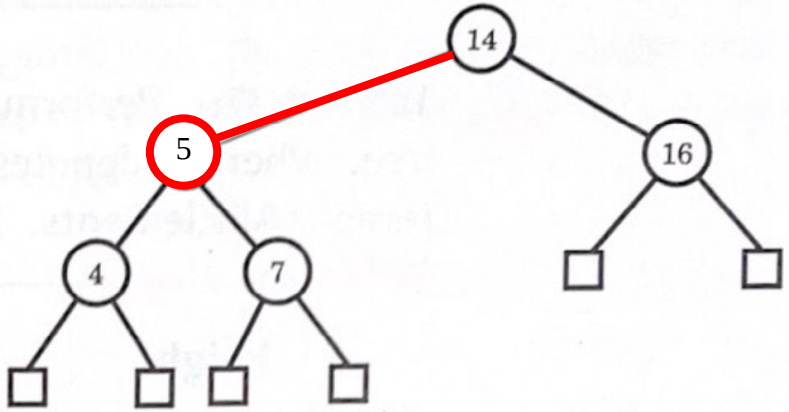
Removal of 17



Removal of 18 (double black)

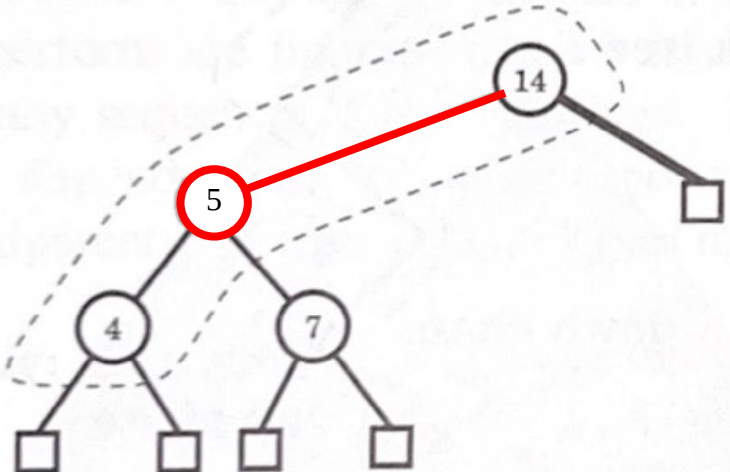


Re-coloring

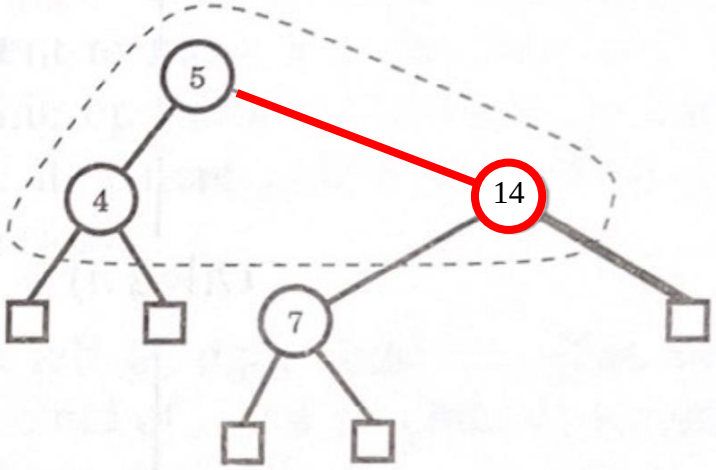


Removal of 15

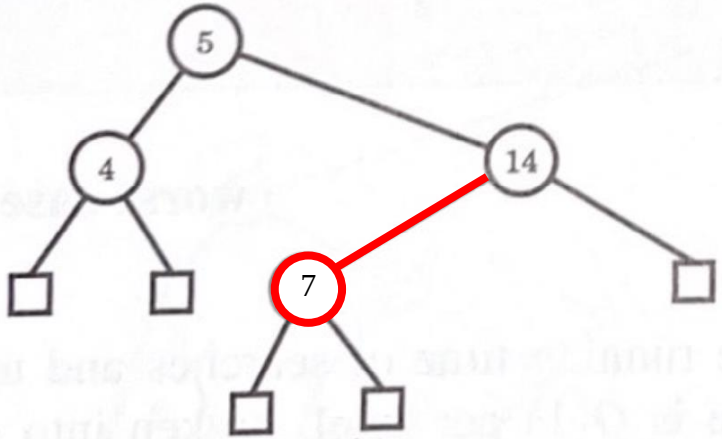
Red-Black Trees: Removal Example



Removal of 16 (double black)



Adjustment



Re-coloring

Extra

Red-Black Tree Reorganization

Insertion	remedy double red	
Red-black tree action	**(2,4) tree action**	**result**
restructuring	change of 4-node representation	double red removed
recoloring	split	double red removed or propagated up

Deletion	remedy double black	
Red-black tree action	**(2,4) tree action**	**result**
restructuring	transfer	double black removed
recoloring	fusion	double black removed or propagated up
adjustment	change of 3-node representation	restructuring or recoloring follows